MIL-M-63160(AR)
25 June 1982
Superseding
(see 6.6)

MILITARY SPECIFICATION

MOTOR BODY
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for rocket motor bodies used on M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-I-6868	Inspection Process, Magnetic Particle
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-410	Non Destructive Testing Personnel Qualification and Certification
MIL-STD-1167	Ammunition Data Cards
MIL-STD-1168	Lot Numbering of Ammunition

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS (see 6.8)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9280261	Magnetic Particle Standard for Motor Body (Envelope dwg)
9287999	Vibra-etch Standard
9289688	Calibration and Operating Instructions for Ultrasonic C-scan Inspection of Rotating Band
9289689	Equipment Tabulation. for Ultrasonic C-scan Inspection of Rotating Band
9299050	Motor Body, 8 Inch M753
9311169	Packing and Marking for Drum, Metal Shipping with Rocket Motor, 8 inch
9325966	preparation for Shipping Rocket Motor
9327972	Motor Body 8 Inch: Hydrostatic Test
9327981	Hydrostatic Test Assembly
9328316	Forging, Motor Body
9329593	Motor Body, 8 Inch, Ballistic Test Evaluation
9329596	Ballistic Test Evaluation Projectile, 8 Inch, PXR6280

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E-8	Tension Testing of Metallic Materials
ASTM E-18	Tests for Rockwell Hardness of Metallic Materials

(Application for copies of ASTM publicatins should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material and forgings. The contractor shall provide objective evidence (see 6.4) that materials and forgings are in accordance with the applicable drawings and specifications.

3.2 Construction. The motor body shall comply with all requirements specified on drawing (dwg. 9299050, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein. Motor body forgings shall comply with all requirements of dwg. 9328316 and all applicable specifications.

3.3 Weld overlay band

3.3.1 Base metal penetration. There shall be penetration of the base metal at the weld interface in the form of arc cratering such that a proper metallic bond is achieved between the copper and base metal.

3.3.1.1 Minimum base weld penetration. Except at the starting weld zone there shall be arc cratering .015 inch deep over a minimum of 90 percent of the entire surface beneath the rotating band. The applicable width for this requirement is the 2.00 + .06 dimension of the rotating

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band seat (dwg. 9327972). The starting weld zone, which is defined as a zone plus or minus one half inch from the centerline of the starting pass, shall have a minimum of 25 percent of the surface covered by arc craters. For the purpose of defining magnitudes of base metal penetration, the value of .015 inch deep shall be used as a guide to preparation of approved visual standards and shall not be literally interpreted as applying to each arc crater. Standards for arc craters shall be prepared and approved and testing shall be performed as specified in 4.4.3.4.3.1.

3.3.1.2 Maximum base metal penetration. There shall be no evidence of penetration (except where the starting weld pass is made) when the base metal is machined to 7.840 diameter minimum. There shall be no evidence of penetration at the starting weld pass when the base metal is machined to 7.770 diameter minimum. Testing shall be performed as specified in 4.4.3.4.2.2 and 4.4.3.4.2.4.

3.3.2 Chemical requirements. The iron content of copper rotating bands shall not exceed 3.0 percent when tested as specified in 4.4.3.4.2.1.

3.3.3 Rotating band cracks. There shall be no evidence of cracks in the finish machined rotating band. In addition, there shall be no evidence of cracks in the base metal beneath the rotating band at the diameters specified in 4.4.3.4.2.3 and 4.4.3.4.2.4.

3.4 Motor body metal defects. Except at the starting weld zone, there shall be no cracks, seams, laminations or laps in the motor body as determined by magnetic particle inspection, except that surface indications of cracks no greater than 1/8 inch in greatest dimension are permitted in the rotating band groove aft of the band if the samples examined in accordance with the weld inspection tests are free of crack indications at the 7.810 inch diameter. Indications of cracks at the starting weld zone must not be present when the weld samples are machined to the 7.770 inch diameter. Non-metallic inclusions will be permitted in the motor body as specified in 3.4.1.

3.4.1 Non-metallic inclusions. Each non-metallic inclusion 1/16 inch and over in length shall be recorded. Non-metallic inclusions shall be no less than 3/8 inch apart. Motor bodies shall conform to a maximum frequency rating of 0.80 and a minimum severity rating of 0.67. The frequency (numbers) and severity (size) of inclusions shall be calculated as follows:

3.4.1.1 Frequency. The number of non-metallic inclusions per motor body shall be totaled. This total shall be

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divided by 520 square inches (surface area of the motor body) to determine the frequency rating per motor body.

3.4.1.2 Severity. The length of each non-metallic inclusion shall be ascertained and multiplied by the appropriate progression factor (see Table I). The sum of these products shall then be divieded by 520 square inches (surface area of the motor body) to determine the severity rating per motor body.

TABLE I. Severity rating of inclusions.

Length of Inclusion	Progression Factor
1/16 to 1/8 inch inclusive	0.5
over 1/8 to 1/4 inch inclusive	1
over 1/4 to 1/2 inch inclusive	2
over 1/2 to 3/4 inch inclusive	4
over 3/4 to 1 inch inclusive	8
over 1 to 1 1/2 inch inclusive	16

3.5 Hydrostatic pressure. Motor bodies shall withstand an internal hydrostatic pressure in accordance with the requirements specified on drawing 9327972 without evidence of leakage, cracks, flaws or ruptures. There shall be no permanent distortion of the motor body beyond the dimensional requirements indicated on dwgs. 9327972 and 9299050. Testing shall be performed in accordance with 4.5.1.

3.6 Rotating band bond. Ultrasonic C-scan inspection shall be utilized to inspect for quality of bond of the rotating band.

3.6.1 Accept-reject criteria. A Motor body shall be classified defective if C-scan inspection shows any unbended area whose maximum linear dimension exceeds one inch (1") except as listed below:

1. A vertical unbend of length the entire width of the band, whose average width is less than 1/2 inch and whose maximum width is less than 3/4 inch shall be acceptable (see Figures 1A, 1B).

2. A vertical unbend which does not touch one or both of the edges of the rotating band and whose maximum with is less than 3/4 inch shall be acceptable (see Figure 1C).

3. An unbend area may extend along entire edge of scan, if its width is less than 3/16 inch, disregarding minor bulges (see Figure 2A).

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4. If the width of the edge of band unbended area does not exceed 1/4 inch and the length does not exceed 2 inches, ,it is acceptable. However, if more than one such area exists, the spacing between them shall not be less than 1/4 inch (see Figures 2B, 2C).

5. A mottled series of small unbends whose area of unbond is not 50% (percent) of the area encompassed and whose total area of unbend is less than 10 square inches shall be acceptable (see Figure 3).

3.6.2 Inspection methods and procedures. Ultrasonic C-scan inspection shall be performed in accordance with 4.5.3 using equipment and procedures contained in dwgs. 9289688 and 9289689.

3.6.3 Motor body inside diametier (Prior to ultrasonic inspection). The inside diameter of the motor body prior to ultrasonic inspection shall be of a contractor specified diameter with a tolerance of .015 inch total and a surface finish of 125 RMS or better in the area of the rotating band. Also, the corner radius at the bulkhead shall be such as not to interfere with C-scan inspection in the area of. the rotatlng band.

3.7 Proving ground test. When assembled on PXR6280 Test Projectiles furnished by the acquisition agency and fired at a government proving ground, there shall be no breakup collapse or band separation of the motor body in the gun tube or in flight. Testing shall be performed in accordance with 4.5.5.

3.8 Serialization. The contractor shall assign serial numbers to each item. Serial codes shall be obtained by means of a letter addressed to Commander, US Army Armament Research and Development Command, Dover, N.J. 07801, ATTN: DRDAR-QAN-R. The letter of request shall include:

a. Nomenclature of item as specified on the applicable dwg.

b. Name of contractor

c. Contract number or acquisition request order number

d. Number of serial codes required.

3.9 Inspection records and data cards. Inspection records of all inspections performed on individual motor

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bodies and ammunition data cards prepared in accordance with MIL-STD-1167 shall be provided and stord in 9311169 shipping drum for each motor body.

3.10 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.11 Workmanship. The requirements for workmanship shall be as specified on the applicable drawings, in the applicable specifications and as specified herein.

3.11.1 Surface preparation. Preparation of metal surfaces shall be as specified in the applicable drawings and in the applicable specifications.

3.11.2 Painting and preservation. Painting and preservation shall comply with the requirements of the applicable drawings. All paint and preservative shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.11.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing.

3.11.4 Burrs. No part shall have burrs which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.5 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion of other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own, or any, other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government Government reserves the right to perfrom any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform

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to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When first article inspection is required: (see 3.10) the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Quantity	Description	Drawing
5	Unhanded motor bodies for tests of 4.4.2.1	9327972
4	Semi-finished banded motor bodies for test of 4.4.2.2	9327972
6	Semi-finished banded motor bodies for test of 4.4.2.3	9327972
5	Motor Bodies, M753: finish machined, for tests of 4.4.2.4 and 4.4.2.5	9299050
20	Motor Body, 8 Inch, Ballistic Test	9329593

All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials including packaging and packing shall be obtained from the same source of supply as will be used in regular production standards and test equipment as required, shall be approved by the procuring activity prior. to submission of the first article sample.

4.3.2 Inspections to be performed. See MIL-A-48078 and the first article inspection table herein.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1	Motor Body, 8 Inch, M753	1 OF 4	9299050 and 9327972
			NEXT HIGHER ASSEMBLY: M753 PROJ

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Motor Body, Unbanded (dwg. 9327972).	5			
	Classification of defects			3.1,3.2 3.11	4.4.2.1
	NOTES: 1. Inspections and tests in any subgroup need not be performed in the order indicated. 2. Tests preceded by an asterisk are destructive and shall be performed after all other non-destructive tests in the indicated subgroup.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1 (cont)	Motor Body, 8 Inch, M753	2 OF 4	9299050 and 9327972
			NEXT HIGHER ASSEMBLY: M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR* 100%	REQUIREMENT PARAGRAPH	*PARAGRAPH REFERENCED /INSPECTION METHOD
	Motor Body, Banded and Semi-finished (dwg. 9327972).	10			
	Prior to mag. particle and hydrostatic test.	4			4.4.2.2
	Classification of defects			3.2,3.11	
	Ultrasonic inspection			3.6	
	*Mechanical properties (see 4.3.2.1.1)			3.2	
	*Properties of rotating band (see 4.3.2.1.2)			3.3	
	Prior to finish machining to dwg. 9299050.	6			4.4.2.3
	Classification of defects			3.2,3.11	
	Magnetic particle inspection			3.4	
	Hydrostatic pressure test			3.5	

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.3.2.1 (cont)	Motor Body, 8 Inch, M753	3 OF 4	9299050 and 9327972
			NEXT HIGHER ASSEMBLY: M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Motor Body, 8 Inch, M753 (dwgs. 9299050 and 9327972).	5			
	Finish machined				
	Classification of defects			3.2, 3.11	4.4.2.4 (see note 3)
	Finish machined - Corrosion Preventive Finished				
	Classification of defects			3.2, 3.11	4.4.2.5

NOTES:
3. Following preliminary acceptance in accordance with 4.4.2.4, the contractor shall complete the items and resubmit for the inspections of 4.4.2.5.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.3.2.1 (cont)	TITLE Motor Body, 8 Inch, Ballistic Test Evaluation		SHEET 4 OF 4		DRAWING NUMBER 9329593
					NEXT HIGHER ASSEMBLY 9329596 (PXR Test Proj)
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Motor Body, Ballistic Test Evaluation (see 4.3.2.1.3 and note 4)	20		3.2,3.7	4.5.5

NOTES:
4. First article shall not be approved until the results of the Proving Ground test of 4.5.5 are evaluated and approved by NSD, PAD (See 6.5).

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4.3.2.1.1 Mechanical properties of motor body. In lieu of the mechanical property screening tests of 4.4.3.3, first article testing of the mechanical properties of motor bodies shall consist of testing two items for the hardness, tensile strength and elongation specified on the dwg. Test methods, location of hardness readings and preparation of tensile strength and elongation specimens shall be in accordance with 4.4.3.3, as applicable. If any motor body fails to meet the hardness, tensile strength or elongation requirements of the dwg., the first article sample shall be rejected.

4.3.2.1.2 Properties of rotating band. In lieu of the rotating band tests on highest hardness and lowest hardness bands of 4.4.3.4, first article testing of the rotating band properties shall consist of testing two items for all the characteristics of 3.3 and 3.4 in accordance with 4.4.3.4.2.1 thru 4 .4.3.4.2.4 and 4.4.3.4.3.1, as applicable. Items shall be visually examined for cracks as specified in 4.4.3.4.2 and 4.4.3.4.3. If any motor body fails to meet the rotating band property requirements, the first article sample shall be rejected.

4.3.2.1.3 Proving ground tests. First article proving ground tests shall be performed on twenty motor bodies assembled on PXR6280 Test Vehicles provided by the acquiring agency. Prior to submission to the acquiring agency for loading and assembly on test projectiles, the contractor shall perform and receive preliminary acceptance of all of the applicable non-destructive tests normally performed on motor bodies in accordance with 4.4.2.1 thru 4.4.2.4 except that the requirements of dwg. 9329593 shall apply in lieu of dwg. 9327972. If any motor body fails to meet the applicable requirements of 4.4.2.1 thru 4.4.2.4 and dwg. 9329593 or the requirements of 3.7, the first article sample shall be rejected.

4.3.3 Rejection See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078: In addition to these provisions the lot formation provisions of 4.4.1.1 thru 4.4.1.2 shall also apply.

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4.4.1.1 Weld lot. Weld. lots are defined as inspection lots of motor bodies which have weld overlays deposited by one unchanged process with no changes of equipment or setting. In addition, weld inspection lots shall not exceed 800 components.

4.4.1.2 Heat treatment inspection lot. A heat treatment inspection lot shall not exceed one forging inspection lot or 500 components, whichever is smallest. Whenever a break in heat treating occurs or a furnace is shut down for any length of time the heat treatment inspection lot will be terminated and a new lot shall be begun. Additionally, heat treatment inspection lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into larger heat treatmnent inspection lots after qualifying with a minimum of five individual batches provided qualifying procedures are submitted to and approved by the Government and provided all the other restrictions detailed above are met. Where controls and procedures are such that batch type furnaces are effectively used in a continuous manner, as determined by the Government, the requirement for qualification with individual batches prior to using the maximum heat treat inspection lot sizes shall be waived. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment inspection lots shall remain grouped together and identified by inspection lot until mechanical property tests have been completed. Components. from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.4.2 Examination. Examination shall be in accordance with MIL-A-48078 and the Quality Conformance Inspection tables herein.

a. Sampling plans. Unless otherwise specified in the Quality Conformance Inspection tables herein, sampling plans for Major and Minor defects shall be in accordance with Tables II and III.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Motor Body: Unbanded	1 OF 1	9327972
			NEXT HIGHER ASSEMBLY 9299050

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Objective evidence for forgings in accordance with dwg. 9328316 and applicable specification		-	3.1	Visual
102.	Diameter of rotating band groove		Table II	3.2	Gage
103.	Runout of rotating band groove		Table II	3.2	Gage
104.	Evidence of poor workmanship		Table II	3.11	Visual
Minor					
201.	Width of Band Groove (min)		Table III	3.2	Gage
202.	Distance from rear face to rotating band groove		Table III	3.2	Gage
203.	Radius of groove (2 places)		Table III	3.2	Gage
204.	Angle of groove (2 places)		Table III	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Motor Body: Banded and Semi-finished, Prior to Magnetic Particle and Hydrostatic Testing		9327972 (see note 1)
			NEXT HIGHER ASSEMBLY 9299050

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Ultrasonic inspection		100%	3.6	4.4.3.6
102.	Ultrasonic inspection stamp missing		100%	3.6	Visual
103.	Rotating band and motor body material		-	3.1	Visual
104.	Mechanical properties of motor body	4.4.3.3	-	3.2	4.4.3.3
105.	Properties of rotating band	Table IV	-	3.3	4.4.3.4
106.	Length of largest cavity		100%	3.2	Gage
107.	Overall length		100%	3.2	Gage
108.	Pitch diameter of largest thread		100%	3.2	Gage
109.	Outside diameter of motor body (2 places)		100%	3.2	Gage
110.	Inside diameter of largest cavity		100%	3.2	Gage
111.	Inside diameter of small cavity adjacent to small end of exit cone		100%	3.2	Gage
112.	Thickness of bulkhead		100%	3.2	Gage

Note: 1. Whenever true position is specified on dwgs. and is to be measured on CORDAX (or equivalent) automatic measuring equipment, measurements shall be made at a minimum of six locations around the diameter and three locations along the length.

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2 (cont)	Motor Body: Banded and Semi-finished, Prior to Magnetic Particle and Hydrostatic Testing	2 OF 2	9327972 (see note 1)
			NEXT HIGHER ASSEMBLY 9299050

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
113.	Radius at bulkhead end in largest cavity (including visual for proper blend)		100%	3.2	Gage/Visual
114.	Bulkhead/boattail radius		100%	3.2	Gage
115.	Bulkhead/delay area radius		100%	3.2	Gage
116.	Location of small basic diameter in exit cone		100%	3.2	Gage
117.	Location of large basic diameter in exit cone		100%	3.2	Gage
118.	Evidence of poor workmanship		100%	3.11	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Motor Body: Prior to Finish Machining	1 OF 1	9327972
			NEXT HIGHER ASSEMBLY 9299050

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Hydrostatic pressure test		100%	3.5	4.4.3.1
2.	Magnetic particle inspection		100%	3.4	4.4.3.2
Major					
101.	Hydrostatic test stamp		100%	3.2	Visual
102.	Magnetic particle inspection stamp		100%	3.2	Visual
103.	Location of porthole		100%	3.2	Gage
104.	Inside diameter of thru hole of porthole		100%	3.2	Gage
105.	Pitch diameter of porthole thread		100%	3.2	Gage
106.	Minor diameter of porthole thread		100%	3.2	Gage
107.	Overall depth of porthole		100%	3.2	Gage
108.	Surface Z not grit blasted and prepared in accordance with dwg.		100%	3.2	Visual
109.	Finish all over		100%	3.2	Visual
*110.	Knurl missing or incorrect (see dwg. 9329593)		100%	3.2	Visual
*111.	Outside diameter over bourrelet (max) (see dwg. 9329593)		100%	3.2	Gage
112.	Evidence of poor workmanship		100%	3.11	Visual

NOTES: *These inspections shall apply only to motor bodies for first article ballistic test projectiles for proving ground tests.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Motor Body, 8 Inch, M753: Finish Machined	1 OF 6	9299050 (see note 1)
			NEXT HIGHER ASSEMBLY: M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Waviness or non-circumferential (radial) flaws on surface Y (3 diameters).		100%	3.2	4.5.4
2.	Surface finish and roughness profile of surface Y		100%	3.2	4.5.4
Major					
101.	Depth of large cavity		100%	3.2	Gage
102.	Inside diameter adjacent to forward end		100%	3.2	Gage
103.	Runout of inside diameter adjacent to forward end		100%	3.2	Gage
104.	Perpendicularity of inside diameter adjacent to forward end		100%	3.2	Gage
105.	Length of inside diameter adjacent to forward end (max)		100%	3.2	Gage
106.	Radius between surface Y and inside diameter adjacent to forward end (max)		100%	3.2	Gage
107.	Radius between surface Y and large cavity (max)		100%	3.2	Gage

NOTES: 1. See dwg. 9327972 for dimensions preceded by an asterisk. No machining operations shall be performed following these inspections which could alter the applicable dimensional characteristics.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4 (cont)	Motor Body, 8 Inch, M753: Finish Machined	2 OF 6	9299050 (see note 1)
			NEXT HIGHER ASSEMBLY: M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
108.	Radius adjacent to forward face		100%	3.2	Gage
109.	Runout of forward face		100%	3.2	Gage
110.	Perpendicularity of rear face		100%	3.2	Gage
111.	Parallelism of forward and rear faces		100%	3.2	Gage
112.	10.58 overall length		100%	3.2	Gage
*113.	Location of 7.07 small basic outside diameter of boattail		100%	3.2	Gage
*114.	Runout of 7.07 small basic outside diameter of boattail		100%	3.2	Gage
*115.	Location of 7.92 large basic outside diameter of boattail		100%	3.2	Gage
*116.	Runout of 7.92 large basic outside diameter of boattail		100%	3.2	Gage
*117.	7.970 outside diameter of motor body		100%	3.2	Gage
*118.	Runout of 7.970 outside diameter at rear of motor body		100%	3.2	Gage
*119.	2.64 outside diameter of body at large end of exit cone		100%	3.2	Gage
*120.	Runout of 2.64 outside diameter of body at large end of exit cone		100%	3.2	Gage
*121.	8.195 outside diameter of rear crest of rotating band (3 places)		100%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4 (cont)	Motor Body, 8 Inch, M753: Finish Machined	3 OF 6	9299050 (see note 1)
			NEXT HIGHER ASSEMBLY: M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
*122.	Runout of 8.195 outside diameter of rotating band (3 places)		100%	3.2	Gage
*123.	8.162 outside diameter of second crest of rotating band		100%	3.2	Gage
*124.	Runout of 8.162 outside diameter of rotating band on second crest		100%	3.2	Gage
*125.	Runout of 8.1519 outside diameter of forward band of rotating band		100%	3.2	Gage
*126.	1.626 inside diameter of small cavity adjacent to small end of exit cone		100%	3.2	Gage
*127.	Runout of 1.626 inside diameter of small cavity adjacent to small end of exit cone		100%	3.2	Gage
*128.	2.245 inside diameter of section between groove and shoulder within exit cone		100%	3.2	Gage
*129.	Runout of 2.245 diameter of section between groove and shoulder exit cone		100%	3.2	Gage
*130.	Inside diameter at rear end of boattail		100%	3.2	Gage
*131.	Runout of inside diameter at rear end of boattail		100%	3.2	Gage
*132.	Full length taper of inside diameter of boattail		100%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4 (cont)	Motor Body, 8 Inch, M753: Finish Machined	4 OF 6	9399050 (see note 1)
			NEXT HIGHER ASSEMBLY: M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR /100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
*133.	Length of three rear band crests at basic location (3 places)		100%	3.2	Gage
*134.	Length of two forward band crests at basic location (2 places)		100%	3.2	Gage
*135.	Width of rotating band		100%	3.2	Gage
*136.	Angle of forward crest of rotating band		100%	3.2	Gage
*137.	Location of basic outside diameter on forward band crest		100%	3.2	Gage
*138.	Outside diameter of rotating band at base of crest (4 places)		100%	3.2	Gage
*139.	Distance from rear face of band to basic dimension on each crest of band (5 places)		100%	3.2	Gage
*140.	Runout of largest cavity		100%	3.2	Gage
*141.	Distance from rear face to rear of rotating band		100%	3.2	Gage
*142.	Outside diameter of band seat adjacent to rotating band		100%	3.2	Gage
*143.	Length of small cavity adjacent to small end of exit cone		100%	3.2	Gage

NOTES:

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FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4 (cont)	Motor Body, 8 Inch, M753: Finish Machined	5 OF 6	9399050 (see note 1)
			NEXT HIGHER ASSEMBLY: M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
*144.	Runout of shoulder at small end of exit cone with diameter of small cavity at small end of exit cone		100%	3.2	Gage
*145.	Distance from cavity at small end of exit cone to groove in exit cone		100%	3.2	Gage
*146.	Distance from rear face of exit cone to flat face at bottom of cavity		100%	3.2	Gage
*147.	Width of groove in large end of exit cone		100%	3.2	Gage
*148.	Depth of groove in large end of exit cone		100%	3.2	Gage
*149.	Location of groove in large end of exit cone		100%	3.2	Gage
*150.	Slots missing, improper, insuffucient or not properly located (8 places)		100%	3.2	Visual
*151.	Surface finish of exit cone improper		100%	3.2	Gage
*152.	Radius at shoulder within large end of exit cone missing or improper		100%	3.2	Visual
*153.	Radius in small cavity adjacent to small end of exit cone improper or missing		100%	3.2	Visual
*154.	Surface finish inside of boattail				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.4 (cont)	Motor Body, 8 Inch, M753: Finish Machined		6 OF 6	9399050 (see note 1)
				NEXT HIGHER ASSEMBLY
				M753 PROJ.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
*155.	Width of band seat adjacent to rotating band		100%	3.2	Gage
*156	Runout of small basic diameter in exit cone		100%	3.2	Gage
*157.	Runout of large basic diameter in exit cone		100%	3.2	Gage
*158.	Profile of rotating band including any damage to crest		100%	3.2	Gage
*159.	Depth of porthole thread (min)		100%	3.2	Gage
*160.	Angle at bottom of porthole		100%	3.2	Gage
*161.	Finish at bottom of porthole		100%	3.2	Gage
*162.	Porthole free of burrs, slivers, feather edges and foreign material (see note 2)		100%	3.2	Visual/Manual
*163.	Rotating band stress relief		-	3.2	4.4.3.5
164.	Surface finish on radius adjacent to forward face		100%	3.2	Visual
165.	Surface finish on inside diameter adjacent to forward face		100%	3.2	Visual
166.	Surface finish on forward face		100%	3.2	Visual
167.	Evidence of poor workmanship		100%	3.11	Visual

NOTES: 2. Burrs, slivers, feather edges and any other defects shall be removed with an approved deburring tool and all foreign matter removed by an approved cleaning method.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Motor Body, 8 Inch, M753: Finish Machined, Corrosion Prevented Finished	1 OF 1	9299050
			NEXT HIGHER ASSEMBLY: M753 Proj

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Any imperfection of forward crest of band		100%	3.2	Visual
102.	Surfaces to be protected by preservative not clean and free of rust, contamination and foreign matter immediately prior to application of corrosion preventive compound.		100%	3.2	Visual
103.	Corrosion preventive compound, missing or bare spots present exposing base material (see dwg. 9325966)		100%	3.2	Visual
104.	Corrosion preventive compound not properly dried (see dwg. 9325966)		100%	3.2	Record
105.	Corrosion preventive compound material		-	3.1	Visual
106.	Evidence of poor workmanship		100%	3.11	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Calibration Standard, Ultrasonic	1 OF 1	See Note Below
			NEXT HIGHER ASSEMBLY: NA

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Inside diameter in area of rotating band to specified contractor diameter		100%	3.6.3	Gage
102.	Surface finish in area of rotating band		100%	3.6.3	Visual
103.	Corner radius of bulkhead improper or protruding into area of rotating band		100%	3.6.3	Gage
104.	Evidence of poor workmanship		100%	3.11	Visual

NOTES: These inspections associated with preparation of Ultrasonic Calibration Standards and performance of ultrasonic inspections (see 4.4.3.6.1 and 4.5.3.1).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Packaging	1 OF 2	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101.	Porthole setscrew improper or missing		100%	5.0	Visual
102.	Improper surface preparation		100%	5.0	Visual
103.	Improper or missing rust preventive compound or protective wrap		100%	5.0	Visual
104.	Any protective closure improper or missing				
105.	Grommet assembly missing or not secured over rotating band		100%	5.0	Visual/Manual
106.	Motor body not inclosed in barrier material or bag		100%	5.0	Visual
107.	Barrier material not sealed or taped		100%	5.0	Visual
108.	Any support or cushion missing from packed container assembly		100%	5.0	Visual
109.	Motor body does not fit snugly in support cavities		100%	5.0	Visual
110.	Filler material insufficient to provide firm pack		100%	5.0	Visual/Manual
111.	Inspection records and data cards not inclosed in container		100%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7 (cont)	Packaging	2 OF 2	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major (cont)					
112.	Container marking missing, incorrect or illegible		100%	5.0	Visual
113.	Evidence of poor workmanship		100%	-	Visual

NOTES:

DD Form 2567 Feb 74 Which is Obsolete

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Type II. Sampling plan for major defects.

LQ = .01 Acceptance No = 0 Consumers Risk = .05

Lot Size	Sample Size
1-72	all
73	72
74	73
75	74
76	75
77	76
78	77
79	78
80	79
81-82	80
83	81
84	82
85	83
86	84
87	85
88-89	86
90	87
91	88
92	89
93	90
94-95	91
96	92
97	93
98	94
99-100	95
101	96
102	97
103	98
104-105	99
106	100
107	101
108-109	102
110	103
111	104
112-113	105
114	106
115	107
116-117	108
118	109
119	110
120-121	111
122	112
123-124	113
125	114
126-127	115
128	116
129-130	117
131	118
132-133	119
134	120
135-136	121
137	122
138-139	123
140-141	124
142	125
143-144	126
145	127
146-147	128
148-149	129
150	130
151-152	131
153-154	132
155-156	133
157	134
158-159	135
160-161	136
162-163	137
164-165	138
166-167	139
168	140
169-170	141
171-172	142
173-174	143
175-176	144
177-178	145
179-180	146
181-182	147
183-184	148
185-187	149
188-189	150
190-191	151
192-193	152
194-195	153
196-197	154

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Table II. Sampling plan for major defects-cont.

LQ = .01 Acceptance No = 0 Consumers Risk = .05

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
198-200	155	330-333	197	638-649	239
201-202	156	334-338	198	650-662	240
203-204	157	339-342	199	663-676	241
205-207	158	343-347	200	667-790	242
208-209	159	348-352	201	691-704	243
210-212	160	353-356	202	705-719	244
213-214	161	357-361	203	720-735	245
215-217	162	362-366	204	736-751	246
218-219	163	367-372	205	752-768	247
220-222	164	373-377	206	769-785	248
223-224	165	378-382	207	786-804	249
225-227	166	383-388	208	805-823	250
228-230	167	389-393	209	824-842	251
231-232	168	394-399	210	843-863	252
233-235	169	400-405	211	864-884	253
236-238	170	406-411	212	885-907	254
239-241	171	412-417	213	908-930	255
242-243	172	418-424	214	931-955	256
244-246	173	425-430	215	956-981	257
247-249	174	431-437	216	982-1008	258
250-252	175	438-443	217	1009-1036	259
253-255	176	444-450	218	1037-1066	260
256-259	177	451-457	219	1067-1098	261
260-262	178	458-465	220	1099-1131	262
263-265	179	466-472	221	1132-1166	263
266-268	180	473-480	222	1167-1204	264
269-271	181	481-488	223	1205-1243	265
272-275	182	489-496	224	1244-1285	266
276-278	183	497-504	225	1286-1330	267
279-282	184	505-513	226	1331-1378	268
283-285	185	514-521	227	1379-1429	269
286-289	186	522-530	228	1430-1483	270
290-293	187	531-540	229	1484-1542	271
294-296	188	541-549	230	1543-1605	272
297-300	189	550-559	231	1606-1673	273
301-304	190	560-569	232	1674-1747	274
305-308	191	570-579	233	1748-1827	275
309-312	192	580-590	234	1828-1914	276
313-316	193	591-601	235	1915-2010	277
317-320	194	602-613	236	2011-2115	278
321-324	195	614-625	237	2116-2232	279
325-329	196	626-637	238	2233-2361	280
				over 2361	298

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Table III. Sampling plan for minor defects.

LQ = .05 Acceptance No = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	35-36	26	98-112	37
17	16	37-40	27	113-131	38
18	17	41-43	28	132-156	39
19-20	18	44-47	29	157-191	40
21	19	48-51	30	192-244	41
22-23	20	52-56	31	245-334	42
24-25	21	57-62	32	335-519	43
26-27	22	63-69	33	520-1120	44
28-29	23	70-76	34	over 1120	45
30-31	24	77-86	35		
32-34	25	87-97	36		

4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables and herein.

4.4.3.1 Hydrostatic pressure. This test shall be performed 100% in accordance with 4.5.1. If any motor body fails to meet the applicable pressure requirements, that motor body shall be rejected.

4.4.3.2 Magnetic particle. This test shall be conducted 100 percent. Hydrostatic testing as required shall be conducted before this test. Each banded finished machined motor body shall be magnetic particle inspected in accordance with 4.5.2. If any motor body fails to meet the requirements of 3.4, that motor body shall be rejected.

4.4.3.3 Mechanical properties of motor body.

4.4.3.3.1 Hardness. Each banded motor body from a heat treated lot shall be identified and subjected to a hardness test on surface "X" (see dwg. 9327972). Three (3) hardness readings shall be taken approximately 120 degrees apart. The test shall be conducted in accordance with ASTM Method E18. The 3 hardness readings thus obtained shall be averaged and used as the basis for selecting the tension test samples of paragraph 4.4.3.3.2.

4.4.3.3.2 Tension test. Two banded motor bodies from the same heat treated inspection lot shall be selected for

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this test (one motor body from the highest hardness reading and one motor body from the lowest hardness reading). Tests may be performed on motor bodies which have been banded, heat treated and stress relieved but prior to finish machining in accordance with dwg. 9327972. From each of these motor bodies four specimens shall be tested to, determine compliance with yield strength and elongation requirements as follows:

a. One from the sidewall directly under the rotating band in a longitudinal direction.

b. One from the sidewall as close to the open end as possible in a longitudinal direction.

c. One from the bulkhead (heavy wall at the closed end which is perpendicular to the axis of the part). This test bar shall-be taken from the center of the. wall such that its axis is on a radial line perpendicular to the longitudinal center line of the part.

d. One from the sidewall in the boattail area as close to the open end as possible in a longitudinal direction.

Testing shall be-conducted in accordance with American Society for Testing and Materials, Method E8 with the largest sub-standard size round specimen possible. If any specimen fails to meet the requirements of the drawing, the heat treat lot shall be rejected subject to retest as specified in 4.4.3.3.3.

4.4.3.3.3 Retest. If the high hardness motor body. fails the yield strength requirement, the entire heat treatment inspection lot shall be rejected. In the event the lowest hardness motor body fails to meet the yield strength requirement, higher. hardness values may be selected and tested in accordance with 4.4.3.3.2 in order to establish the acceptable minimum hardness. Once this minimum hardness has been established, all motor bodies with hardness values below the new minimum shall be rejected. If the low hardness motor body fails to meet the elongation requirement, the entire heat treatment inspection lot shall be rejected. In the event the highest hardness motor body fails to meet the elongation requirement, lower hardness values may be selected and-tested in accordance with 4.4.3.3.2 in order to establish the acceptable maximum hardness. Once this maximum hardness his been established, all motor bodies with hardness values above the new maximum shall be rejected.

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4.4.3.4 Rotating band. The following table shall be used to select samples for testing the rotating band.

Table IV. Rotating band hardness sample plan

Weld Lot Size	Sample Size	Selection of Samples From Highest Average Hardness Reading	Selection of Samples From Lowest Average Hardness Reading
1-110	4	2	2
111-300	6	3	3
301-800	8	4	4

4.4.3.4.1 Rotating band hardness. This test shall be conducted 100 percent on all finish machined rotating bands after the final heat treatment of the motor body. The motor body shall be identified and 3 readings taken in the center of the band and spaced equidistant around the periphery. The test shall be conducted in accordance with ASTM Method E18 using the Rockwell E scale. In the event there is more than a 4 point spread in the 3 readings an additional 3 readings shall be taken. The total of 6 hardness readings shall be spaced equidistant apart and the 6 readings averaged for the hardness of that motor body band. The average hardness reading shall be ordered and used as the basis for selecting rotating band tests specified below.

4.4.3.4.2 Highest average hardness test sequence. Samples from the highest average hardness readings shall sequentially tested as specified in 4.4.3.4.2.1 through 4.4.3.4.2.4 with rejection as specified in 4.4.3.4.2.5. Prior to testing the samples shall be visually examined for surface cracks and if any sample fails to comply with 3.3.3 the weld inspection lot shall be rejected.

4.4.3.4.2.1 Chemical analysis (see 3.3.2). A sample shall be obtained from each motor body by machining the band from .020 inch above the finished band diameter down to the finished band diameter. All chips shall be collected from each motor body and a minimum of 25 grams shall be randomly selected from the collected chips. The specimens obtained from each band shall be dissolved and the analysis made from the individual solutions. Chemical analysis shall be conducted in accordance with Federal Test Method Standard No. 151 except the samples may not be cleaned with a magnet. If any sample fails to meet the requirement of 3.3.2 the lot shall be rejected, subject to retest as specified in 4.4.3.4.2.5.

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4.4.3.4.2.2 Maximum penetration (other than starting pass) (see 3.3.1.2). After completion of the chemical analysis test, the motor body rotating band groove shall be machined to a minimum diameter of 7.840 inch. The surface shall be free of arc craters except at the first weld pass across the band groove.

4.4.3.4.2.3 Rotating band cracks (see 3.3.3). Upon completion of the 7.840 minimum diameter inspection, the motor body beneath the rotating band groove shall be machined to a min. 7.810 inch diameter and magnetic particle inspected in accordance with 4.5.2 and MIL-I-6868. The presence of cracks in the area of the starting weld zone shall not be cause for rejection, provided they are. machined off at the 7.770 inch diameter specified in 4.4.3.4.2.4. Controls shall be maintained to assure proper identification of the Starting weld zone after machining.

4.4.3.4.2.4 Maximum penetration (including starting pass) (See 3.3.1.2). After testing as specified in 4.4.3.4.2.3 the motor body beneath the rotating band shall be machined to a minimum diameter of 7.770 inch. This machined surface shall be free of any arc craters and cracks. If the 7.810 diameter is free of cracks then magnetic particle inspection at this 7.770 inch diameter is not necessary. If it is necessary to inspect this 7.770 inch diameter for cracks, then magnetic particle inspection shall be used in accordance with 4.5.2 and MIL-I-6868.

4.4.3.4.2.5 Rejection. If any test in the highest average hardness sequence fails to meet the requirements in section 3.3, a new sample equal in quantity to the original shall be selected from the next highest average band hardness reading. Selection of samples shall continue until all tests meet the applicable requirement and an acceptable high average hardness range is established. All units which fall outside the acceptable hardness range shall be rejected.

4.4.3.4.3 Lowest average hardness test. Samples from the lowest average hardness reading shall be tested as specified in 4.4.3.4.3. with rejection as specified in 4.4.3.4.3.2. Prior to testing the samples shall be visually examined for surface cracks and if any sample fails to comply with 3.3.3 the weld inspection lot shall be rejected.

4.4.3.4.3.1 Minimum penetration (see 3.3.1.1). The copper rotating band shall be dissolved by any method which will not damage the weld interface surface. The surface

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shall be visually observed for minimum penetration by comparison of the base metal surface with contractor prepared visual standards. The visual standards shall be made from the same material, utilizing the same groove configuration and processes to be used in banding actual motor bodies and shall be approved by the acquiring activity.

NOTE: The rotating band may be machined to a minimum diameter of 7.920 prior to dissolving the remaining copper.

4.4.3.4.3.2 Rejection. If any unit fails the minimum penetration test a new sample equal in quantity to the original shall be selected from the next lowest average band harness reading. Selection of samples shall continue until the sample passes the minimum penetration requirements and an acceptable low hardness range is established. All units which fall outside this acceptable range shall be rejected.

4.4.3.5 Stress relief. The contractor shall comply with the stress relief requirements of dwg 9327972. For verification the contractor shall submit a certificate of compliance that rotating bands have been stress relieved in accordance with the drawing.

4.4.3.6 Ultrasonic inspection.

4.4.3.6.1 Motor body inside diameter prior to ultrasonic inspection. The inside diameter and surface finish shall be measured 100 percent (inside diameter gaged and finish examined visually) in the area of the rotating band for conformance to requirements, specified in 3.6.3 (see 4.4.2.6).

4.4.3.6.2 Ultrasonic inspection. This inspection shall be performed 100 percent in accordance with 4.5.3 after copper deposition, heat treating, and inside diameter machined as specified in 3.6.3 but prior to finish machining of the outside diameter. Failure of the motor body to meet the requirements of 3.6 shall be cause for rejection.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see MIL-A-48078 and 6.3 herein).

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4.5 Test methods.

4.5.1 Hydrostatic pressure test (see 3.5). All motor bodies shall be hydrostatically tested with an approved test fixture. Dwg. 9327981 is referenced as a guide to preparation of a suitable fixture. Motor bodies initially incorporate a threaded joint (dwg. 9327972) to facilitate use of the test fixture. The threaded joint of dwg. 9327972 must subsequently be removed during final machining operations (dwg. 9299050).

4.5.2 Magnetic particle inspection (see 3.4). This inspection shall be in accordance with specification MIL-I-6868 using approved inspection equipment and the continuous, wet, fluorescent or non-fluorescent method. Only circular magnetization induced by means of a central conductor is required provided the inspection process is determined to be adequate each time it is checked with a standard in accordance with dwg. 9280261. For complete motor body inspection in accordance with 4.4.3.2 the suspension shall be applied to all surfaces of the entire motor body. For weld penetration inspection in accordance with 4.4.3.4.2.3 or 4.4.3.4.2.4 only the machined area beneath the rotating band must be covered with suspension. The magnetic solid content of the suspension shall be maintained at 0.2 - 0.4 milliliters (ml) of solids per 100 mls of suspension when oil is used and 0.15 - 0.35 mls of solids per 100 mls suspension when water is used. Three applications of the magnetic field shall be made on the motor body with a minimum of 2500 amperes (2800 preferred) per application. The body shall be rotated 120 degrees after each application. After draining the excess suspension the motor body shall be inspected under ultraviolet light for the presence of defects. . The adequacy of the inspection process shall be checked at the start of daily production and at eight hour intervals thereafter using a standard in accordance with dwg. 9280261. All plugs in the standard shall be clearly defined four a full 360 degrees both inside and out. If the inspection process is determined to be inadequate the condition shall be corrected and the preceding 8 hours production shall be reinspected at the discretion of the Government inspector. The requirement of specification MIL-I-6868 to check system effectiveness by means of a steel ring specimen shall be waived. The operator(s) of the equipment and the inspector(s) making the determination shall be certified in accordance with MIL-STD-410.

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4.5.3 Ultrasonic inspection (see 3.6).

4.5.3.1 Calibration standard. An ultrasonic inspection Standard shall be manufactured by the contractor for use during the ultrasonic inspection. This standard shall be prepared under the instruction of an ARRADCOM representative for approval. It shall be manufactured in the same configuration as the motor bodies to be inspected (i.e., I.D. machined as specified in paragraph 3.6.3, heat treated but prior to finish machining of the O.D.). The copper band shall be deposited in such a way as to assure of an area of unbond of 1 to 2 square inches with an irregular perimeter of marginal bonding. This area shall be of shape which will allow easy detection in change of size or shape of the unbond area with a small change or drift in the setting of the equipment.

Prior to performing the ultrasonic inspection, the standard shall be inspected by the contractor as specified in 4.4.2.6 to determine acceptability for use and then submitted along with a certificate or conformance to ARRADCOM, Dover, New Jersey, 07801, ATTN: DRDAR-QAN-I for approval. The Government shall machine two flat bottom holes and create a series of three c-scans for calibrating equipment. (See dwg. 9289688 and Figure 4.)

In the event the area of unbond does not readily show the effects of equipment drift then a second motor body shall be manufactured. Any change in process or physical characteristics or dimensions in the bond area shall require production of a new standard, as determined by an ARRADCOM Engineer.

4.5.3.2 Calibration and operational procedures and equipment. Prior to initiation of ultrasonic inspection, the contractor shall submit for written approval one copy of his proposed procedures , including description of equipment (contractor design) and operating procedures (see 4.4.4). No change to the procedures shall be made after the original approval without submission and approval of the proposed changes. As a minimum the adequacy of the inspection process shall be checked by running a calibration check on the calibration standard at the start and finish of each production shift or change in operator, as well as at four (4) hour intervals during production. The scan created shall fall within limits of standard scans provided by ARRADCOM. If the inspection process is determined to be inadequate the condition shall be corredcted and all items.

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inspected since the last acceptable inspection check shall be reinspected at the discretion of the Government Representative. A log book shall be maintained by the operator on each ultrasonic system which, as a minimum, documents the processing of standards, number of items inspected, changing of transducers, instrumentations, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

After completion and acceptance of the ultrasonic tests, each acceptable motor body shall be marked in 1.00 high characters the letter "U" using a rubber stamp and then vibra-etched .25 inch high after rinal machining. Location of the marking is to be the same as for the hydrostatic and magnetic particle acceptance marks. Any rejected motor bodies shall be clearly marked as such and removed from the acceptable motor bodies.

4.5.3.3 Operator qualification. The personnel operating the Ultrasonic Inspection Equipment shall be qualified under MIL-STD-410 as an Ultrasonic, Level 1-Special, or under the direct supervision of a person so qualified.

4.5.3.4 Equipment calibration. The pulse echo control unit shall be returned to the manufacturer for an annual maintenance overhaul and alignment. Each transducer used shall be certified annually including an oscilloscope waveform photograph and an energy-frequency spectrum photograph.

4.5.4 Surface waviness. Surface waviness (including average roughness height, roughness profile, etc.) shall be inspected for drawing requirements and results recorded using an approved recording type profilometer.

4.5.5 Proving ground test (see 3.7). These tests shall be performed at a Government proving ground with motor bodies loaded and assembled on an approved test vehicle (PXR6280) by the acquiring agency. The test shall be conducted in accordance with an approved test program. request (TPR) provided to the proving ground commander by the acquiring agency and witnessed and evaluated by the cognizant PAD element of ARRADCOM (see 6.5). The test vehicle shall be fired at 130°F in the rocket-on mode from an M201E1 series cannon using an M188E1 propelling charge conditioned at 145°F (Zone 9) with chamber pressure of

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47000 $^{+1000}_{-1500}$ p.s.i. Interior ballistic information such as pressure-time etc, shall be maintained for each projectile fired. Cameras shall be positioned to verify:

a. Projectile structural integrity

b. Proper projectile assembly including, marking of rocket motor/warhead relative position at muzzle end of cannon.

c. Ignition and burning performance of the delay column.

Exterior ballistic range and deflection performance shall also be recorded. Adequate photographic and instrumentation coverage shall be provided to assure compliance with 3.7. In the event of abnormal behavior other than specified in 3.7, an attempt shall be made to recover and examine the metal parts for evidence of deficiencies in the motor body structural integrity, as opposed to deficiencies in the structural integrity of other test vehicle components, in order to establish the cause of failure. Abnormal behavior shall be defined as failure to maintain structural integrity or to perform to known M650 interior and exterior ballistic characteristics.

5. PACKAGING

5.1 Packaging, packing and marking. Packing, packing and marking shall be in accordance with the applicable requirements of dwgs. 9325966 and 9311169.

6. NOTES

6.1 Intended use. The motor bodies covered by this specification are intended for use in rocket motor/bulkhead assemblies for M753, 8 Inch Rocket-assisted Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Serialization required (see 3.8)

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c. First article inspection when required (see 3.10).

d. Government furnished property, when-required (see 3.7).

6.2.2 Date requirements. When this specification is used in an acquistion which incorporates. a DD1423 Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR7-104.9 (n) (2) are involved and a DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contractor purchase order requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph no.	Data requirements	Applicable DID no.
(a) 3.9	Data cards	DI-L-1410
(b) 3.9	Data feedback (inspection records)	DI-R-1724
(c) 4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions shall be obtained from the Naval publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. A statement (certification), supported by test data that all materials and forgings produced or purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in contractual documents.

6.5 Proving ground tests. TPR preparation, test firing and test evaluation shall be performed by QEB, NSD, PAD (DRDAR-QAN-Q). A short summary of tests results shall be furnished to the Evaluation and Metrology Div., PAD (DRDAR-QAM) within 5 days after completion of the test for inclusion in the first article test report.

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6.6 Supersession data. This specification includes the requirements of Purchase Description AR-PD-4, Revision A dated 7 July 1980.

6.7 Government furnished property. When first article inspection is required, the contracting officer should arrange to have twenty PXR6280 Test Projectiles fabricated, loaded and packed at ARRADCOM for the proving ground tests specified in 3.7 and 4.3.2.1.3.

6.8 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading US Army Armament Research and Development Command may also include dwgs, and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal, dwgs and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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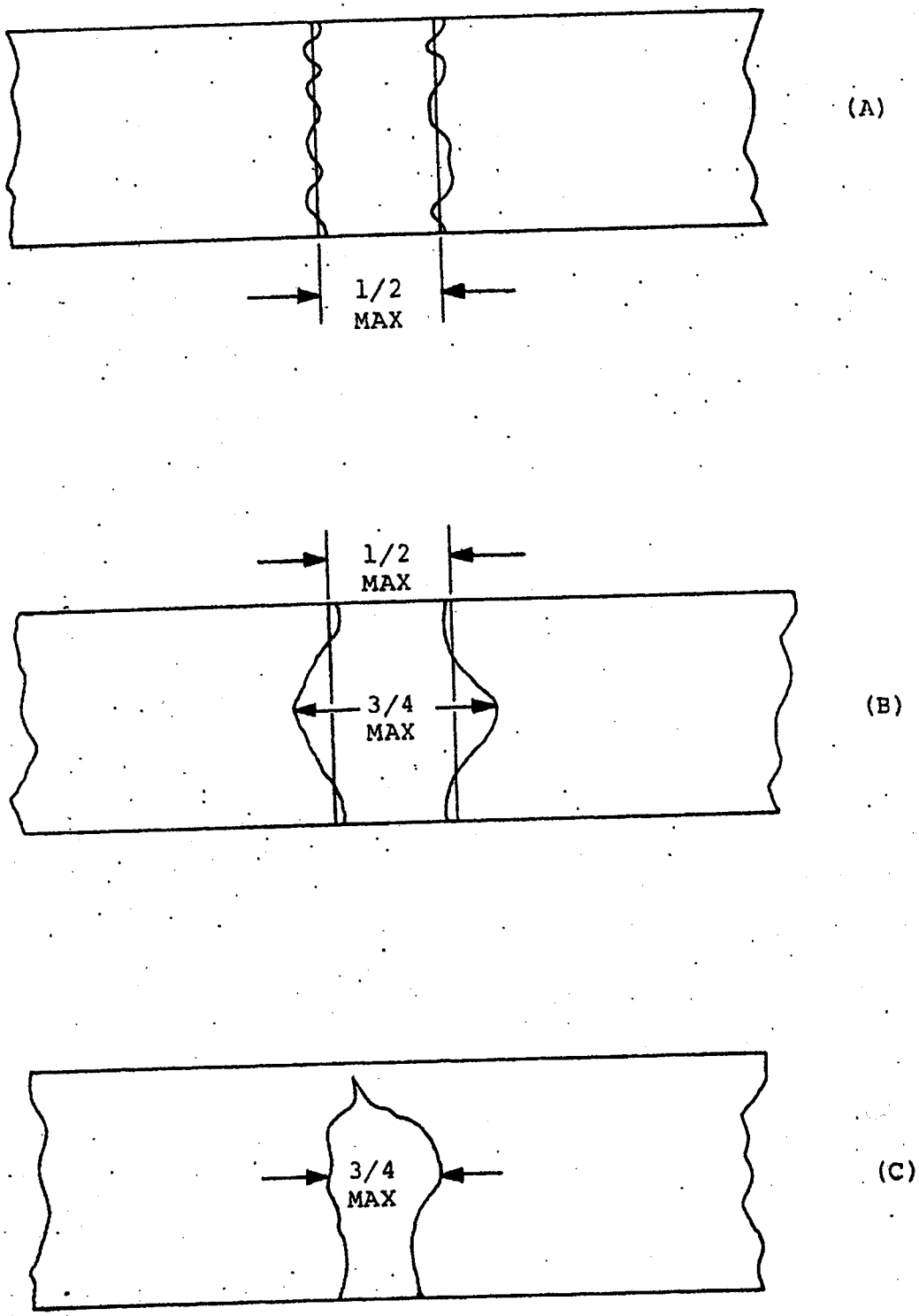


FIGURE 1. Rotating band vertical unbend limits.

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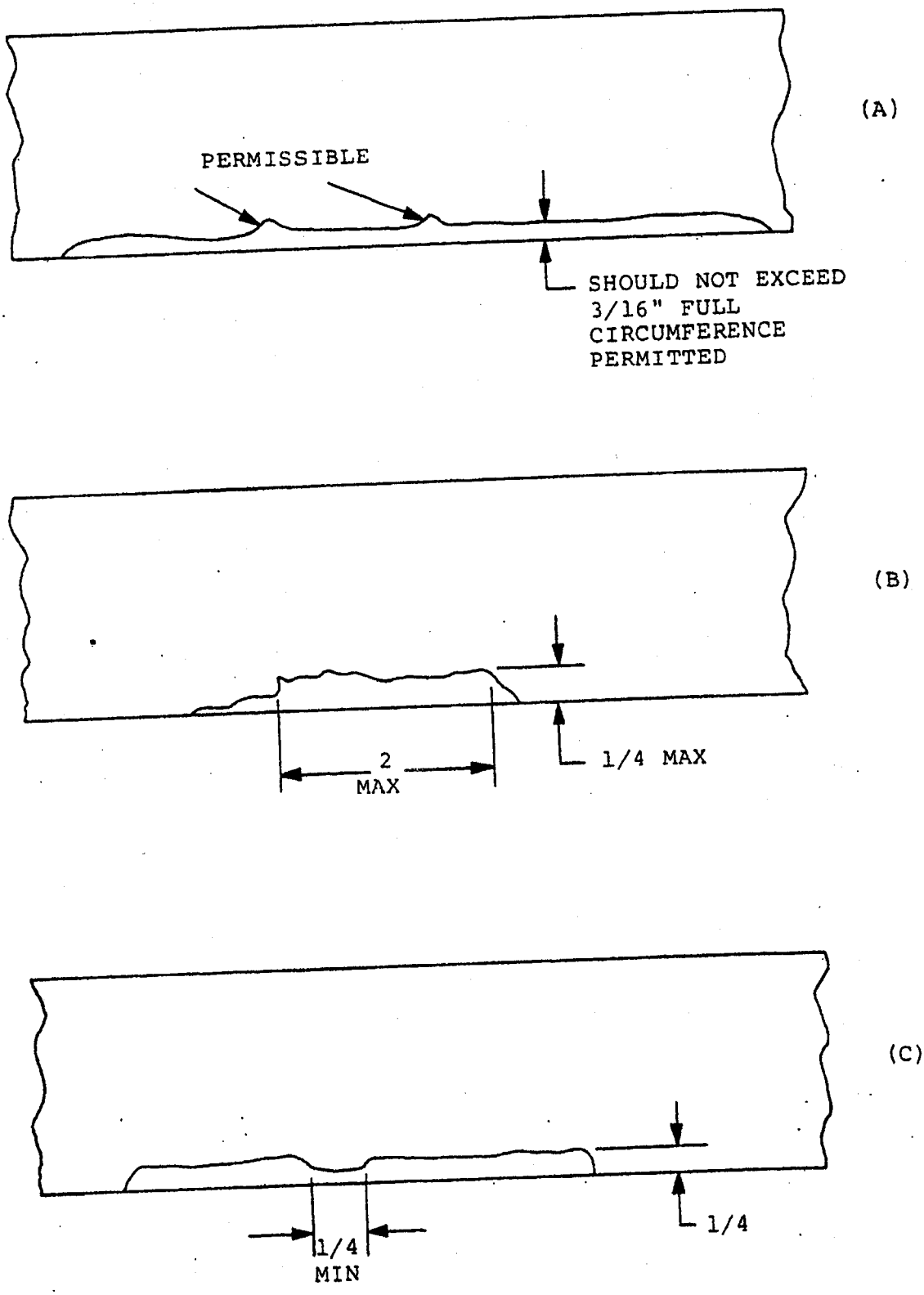


FIGURE 2. Rotating band edge unbonded limits.

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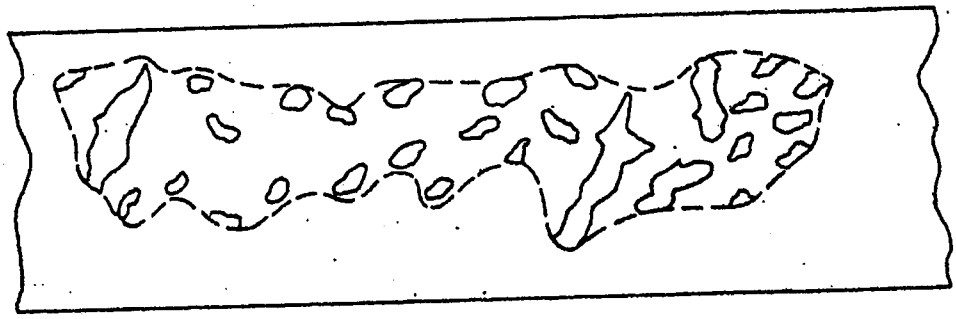


Figure 3. Rotating band mottled unbended area.

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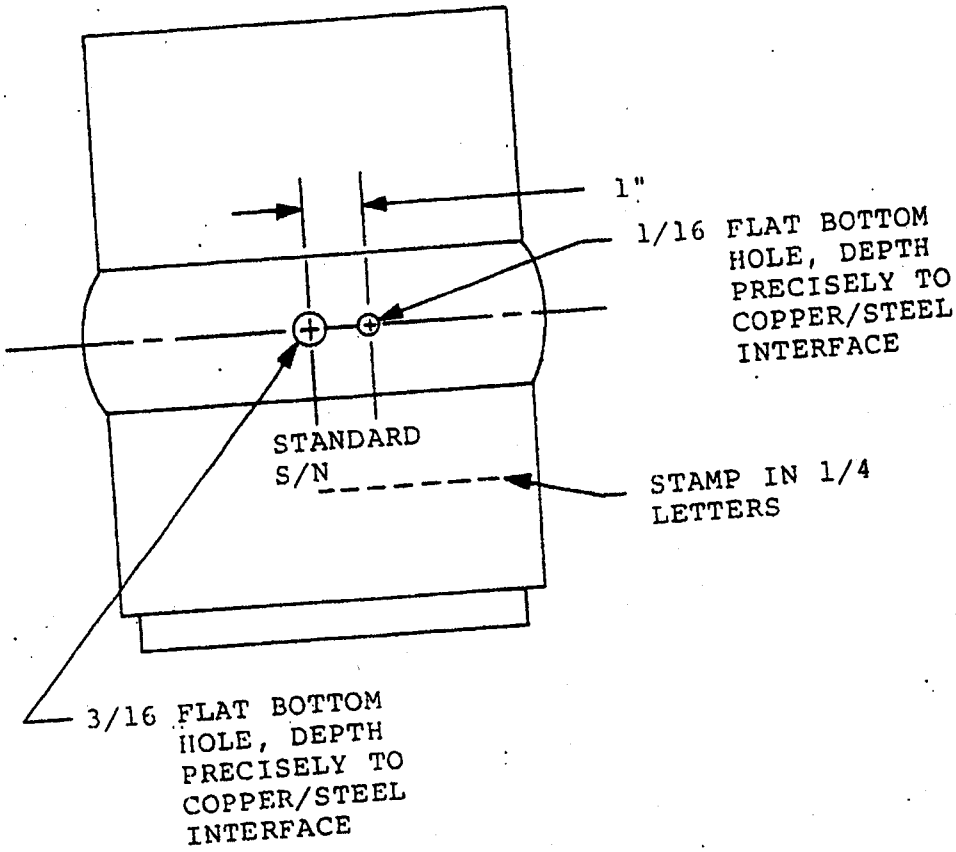


FIGURE 4. Defect standard

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